

Comparing Cinematic Conventions through Emotional Responses in Cinematic VR and Traditional Mediums

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ABSTRACT

This paper compares emotional responses to cinematic conventions in cinematic virtual reality (CVR) versus traditional mediums. We conducted a between-subjects experiment to evaluate viewer experiences across 10 cinematic shot types displayed through three viewing mediums: computer screen, cinema projector, and VR headset (N=15 per condition). Using the Self-Assessment Manikin scale (SAM), we measured pleasure, arousal, and dominance reactions. The statistical analysis showed that CVR elicited significantly higher arousal and lower dominance than traditional formats, indicating enhanced emotional engagement. Overall patterns were found to be similar, but the viewers' emotions were intensified in CVR for shots like close-up and ground-level. Our findings suggest CVR's embodied, interactive qualities altered the impact of cinematic techniques. This comparative study reveals CVR invokes different emotional resonance versus traditional mediums, contributing insights on adapting storytelling strategies for the VR cinematographic experience.

CCS CONCEPTS

• **Computing methodologies** → Computer graphics; Graphics systems and interfaces; Virtual reality.

KEYWORDS

Cinematic virtual reality, Cinematic techniques, Storytelling, Viewer experience

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1 INTRODUCTION

Immersive technologies like cinematic virtual reality (CVR) are emerging as powerful new media for visual storytelling. In traditional cinematic practices, elemental shots are selected and sequenced to establish visual narration. These cinematic techniques are arguably aimed at invoking audience emotions to facilitate storytelling [Arijon, 2013]. However, the unique qualities of CVR, including embodied viewing and interactivity, substantiate a re-examination of cinematic techniques' impact on viewer engagement [Dooley, 2020]. This raises important questions around how traditional visual language translates to VR, and whether new technical principles are needed to leverage CVR's immersive strengths. [Mateer, 2017].

To address this gap, we conducted a comparative study on emotional responses to fundamental cinematic shot sizes and angles in CVR versus traditional medium. Using a between-subjects experiment and SAM, we measured reactions to identical shots across two traditional and a VR viewing conditions. Our findings contribute unique insights into CVR's significance in heightening engagement and altering shot meanings. We discuss implications for adapting cinematic conventions to enable more impactful, emotionally resonant CVR storytelling. This work empirically demonstrates CVR's narrative advantages, while suggesting the potential for exploring a new set of lens language in such a compelling medium.

2 RELATED WORKS

Prior studies have examined various aspects of cinematic experiences in VR, including cognitive and emotional responses to different formats [Ausin-Azofra et al., 2021]. Other works have looked at spatial cognition in relation to different VR interface designs and camera perspectives [Alzayat et al., 2019]. Research has also explored proxemics and interpersonal distance in VR narratives [Keskinen et al., 2019]. Emotional measurement tools like the SAM have been validated for VR contexts [Handayani et al., 2015]. Additionally, Shin developed and tested a VR experience model integrating elements of presence, flow, empathy, and embodiment [Shin, 2018]. However, despite these related attempts, empirical evidence specifically focused on how traditional cinematic techniques translate to VR storytelling remains limited.



Figure 1: Cinematic shots viewed through a Computer Screen, Cinema Projector, and VR headset

3 METHOD

This study implemented a between-subjects experimental design to explore the differential emotional impact of cinematic shot types displayed in traditional 2D medium and VR. The collected data, i.e., the emotional responses, were averaged in accordance with the viewing conditions and further examined with the Analysis of Variance (ANOVA) technique to verify the statistical significance in observed differences.

3.1 Participants

For our experiment, we recruited a total of 45 participants, all of whom were college students in their early twenties, providing a level of comparability across groups in terms of age and educational background. Most of the participants had little to no prior experience with VR. All participants were provided with a detailed explanation of the procedure and subsequently gave their informed consent. They were randomly assigned to three groups of 15 participants for the three corresponding viewing conditions.

3.2 Materials and Apparatus

We utilized a game engine (Unreal Engine 5) as our content creation tool. After thoroughly filtering and selecting from traditional cinematographic techniques, we generated ten representative cinematic shot types defined with visible portions of the human subject in the frame, the distance between the camera and the subject, and the angle of the camera relative to the subject. These techniques included long shot, full shot, mid shot, close shot, close-up, high angle shot, low angle shot, ground-level shot, eye-level shot, and bird's eye shot. These are the most representative and frequently used cinematic techniques in traditional film making. As shown in Figure 1, these generated shots were displayed on a computer screen and a cinema projector, simulating a traditional cinematic viewing environment, while the VR condition was experienced through a high-quality VR headset (Pico 4), immersing the viewer in the virtual scene.

3.3 Experimental Design and Procedure

In our between-subjects design, participants were exposed to their corresponding traditional viewing conditions and VR environment. Under each condition, participants viewed a series of ten rendered

video clips, each featuring a different combination of camera frame shots and angles. For each participant, the shots were presented in a randomized sequence to mitigate ordering effects. During viewing each clip, they rated their emotional reactions to the presented shot using the chosen emotional scale.

3.4 Measures and Data Analysis

We recorded these emotional responses using the SAM, a non-verbal pictorial assessment technique that measures the pleasure, arousal, and dominance associated with a person's affective reaction to various stimuli, with participants rating their emotional responses on a scale from 1 to 9 for each video clip [Bradley and Lang, 1994]. We then conducted an ANOVA to test for significant differences in emotional responses across the different viewing conditions.

4 RESULTS

Based on the measurement of SAM, Figure 2, Figure 3 and Figure 4 present the mean of emotional responses in different viewing medium.

As shown in Figure 2, it's evident that in VR, pleasure levels increase as the shot distance increases and decrease as the shot gets closer, a trend that is more pronounced compared to other mediums. Additionally, when considering camera angles, the emotional response in terms of pleasure in VR is more intense than in the other two mediums.

The bar graph in Figure 3 demonstrates that the VR consistently elicits higher arousal levels across almost all shot types compared to the other two mediums. Moreover, in most cases, the projector's arousal levels surpass those of the screen, implying that larger screen dimensions and wider viewing angles may intensify arousal. We could also observe that the mean arousal in shot size is highly related with proximity especially in CVR. The ground-level shot and close-up shows a significant increase, which stands out compared with others.

The bar in Figure 4 shows that the VR generally induces lower dominance levels for most shot types compared to the other medium. The distribution and structure of mean dominance in these two bar charts closely align with the mean pleasure level shown in Figure 2. This congruence is especially evident in the ground-level shots

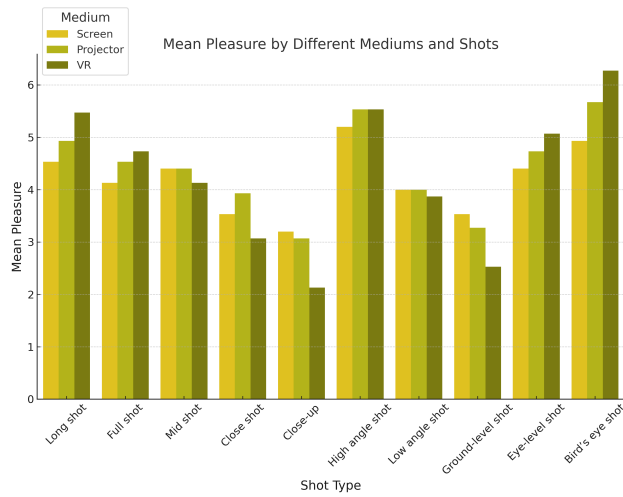


Figure 2: Mean pleasure levels for different shots across mediums.

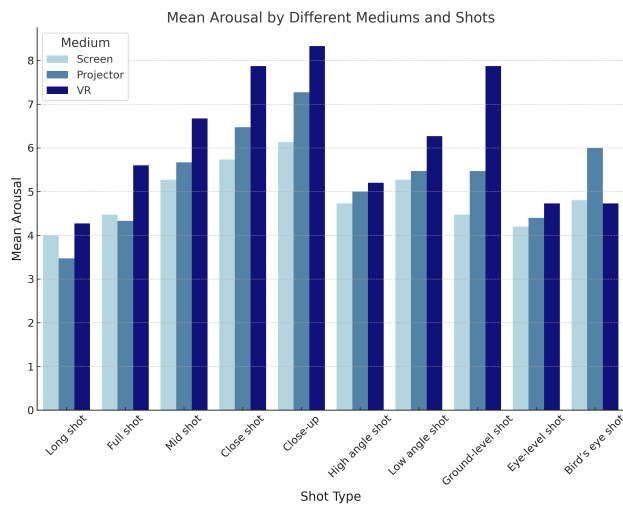


Figure 3: Mean arousal levels for different shots across mediums.

and close-up, where the mean and changes are similar. The ground-level shot and close-up remains exceptional, further confirming its unique characteristics in the context of CVR.

To verify the statistical significance of the different medium impact on emotional response and which groups have significant difference between, we applied the ANOVA, including the post-hoc test.

The results presented in Table 1 indicate the effects of different mediums on all three emotional responses, suggesting the significance (<0.05) at arousal and dominance. To further ascertain the pronounced influence of the VR medium among the groups, we applied a post-hoc analysis.

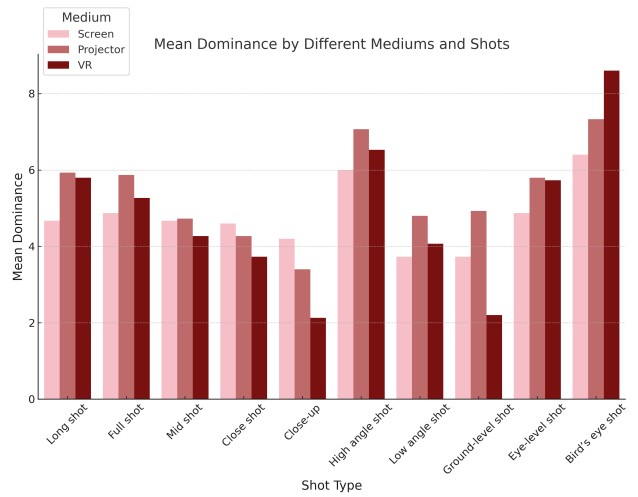


Figure 4: Mean dominance levels for different shots across mediums.

Table 1: The results of a two-way ANOVA, investigating the effects of mediums and shot type, as well as their interaction, on the dependent variables of Pleasure, Arousal, and Dominance.

DV	IV	df	F	Sig.	Par. η^2
Pleasure	Medium	2	0.688	.503	.003
	Shot	9	15.202	.000	.246
	Medium*Shot	18	1.008	.449	.041
Arousal	Medium	2	24.647	.000	.105
	Shot	9	19.159	.000	.291
	Medium*Shot	18	2.427	.001	.094
Dominance	Medium	2	5.534	.004	.026
	Shot	9	22.636	.000	.327
	Medium*Shot	18	2.352	.001	.092

Table 2: The result of a post-hoc test, determine which groups have significant difference between.

DV	Med. (I)	Med. (J)	Mean Diff. (I-J)	Std. Err.	Sig.
Pleasure	VR	Screen	0.09	.188	.620
Pleasure	VR	Projector	-0.13	.188	.502
Arousal	VR	Screen	1.25*	.180	.000
Arousal	VR	Projector	0.80*	.180	.000
Dominance	VR	Screen	0.06	.212	.778
Dominance	VR	Projector	-0.58*	.212	.007

As shown in the Table 2, the VR consistently elicited significantly higher arousal levels compared to both the screen and projector,

suggesting that the medium used can influence feelings of arousal and dominance but has less impact on pleasure levels.

In summary, the results suggest that while the emotional response trends across different shot types are generally similar or three mediums, CVR tends to amplify emotional perceptions. This amplification is particularly noticeable in the case of ground-level shot and close-up.

5 DISCUSSION

Traditional viewing mediums, such as computer screens or cinema projectors, present certain limitations. For instance, when viewers watch content on a computer screen, the size of the screen limits their spatial awareness, thereby challenging their ability to effectively engage with the displayed content. Conversely, cinema projectors offer a larger display and a stronger visual impact, but still restrict viewers to a fixed viewing angle (Figure 1). When participants wore a VR headset, they found that they could engage within the content much easier, establishing a natural sense of distance and size perception within the scene. The freedom of head orientation in CVR also allows participants have micro-interactions with the scene, building deeper spatial and psychological connections.

The cinematic techniques remain efficient and effective in evoke audience emotion in CVR, with the three viewing conditions sharing a similar bar graph structure. However, our results indicates that the medium type significantly influenced arousal and dominance. These findings suggest that traditionally accepted cinematographic techniques may provoke different and more engage emotional response across display mediums, especially in the case of CVR storytelling. The lack of significant variation in pleasure across mediums might depend on the content participants viewed; whether the viewer is pleasant or not in the VR experience is much more personal than other mediums.

In traditional cinematic conventions, ground-level shot is typically used to establish a scene or to offer a unique perspective. However, our experiment revealed a distinct role for these shots in the CVR. Participants often reported feelings of discomfort during ground-level shots, with comments such as "It feels like I'm seeing a giant, I'm being stepped on, it's very uncomfortable." This strong sense of oppression and dominance, akin to crawling on the ground, marks a significant shift in the emotional response. This finding, along with the heightened emotional responses elicited by close-up in CVR, suggests that while some traditional cinematographic techniques can be effective in CVR, others need to be used thoughtfully. Despite these differences, our results indicate that CVR follows a similar structure to the traditional medium in evoking emotional responses but with an enhanced level of intensity. This inspires the unique narrative advantages of CVR, particularly in genres such as horror, war, and documentary, in which the enhanced sense of presence and interactivity can be leveraged to create more immersive and emotionally engaging experiences.

6 CONCLUSION

This comparative study reveals CVR's distinct characteristics as an emerging storytelling medium, relative to traditional viewing formats. By testing fundamental cinematic shot types' impact on

viewer emotions, we demonstrate that identical techniques can elicit significantly different responses across mediums.

Our findings indicate emotional response patterns remain broadly similar, but CVR enhances engagement via heightened arousal and lowered dominance. This implies cinematic conventions may be leveraged to an extent, yet new understandings or techniques need establishment to fully utilize CVR's embodied, interactive nature. Specific shots like ground-level and close-up may take on altered meaning in CVR due to its immersive factors, with implications for emotive genres where presence and interactivity heighten engagement.

While our study provides valuable insights, limitations exist. Our participants sampling strategy serves the purpose for comparing the concerned variables, and statistical significance have been observed. However, it needs further investigations on whether the findings can be applied to wider groups with more varied VR experiences. Future studies could consider participants' familiarity with VR and its effect on the evoked emotional responses. We focused on a limited cinematic technique set, excluding aspects like camera movement. Individual differences in VR experience and motion sickness susceptibility were not considered but could impact CVR experiences. Our video clips utilized a neutral CGI model, which may not reflect actual CVR production content richness.

Future research directions include exploring additional cinematic elements, individual differences, content variety, and CVR narrative structures. By revealing CVR's unique emotional resonance capacities, this comparative work contributes empirical insights on tailoring visual language for more affecting immersive storytelling.

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